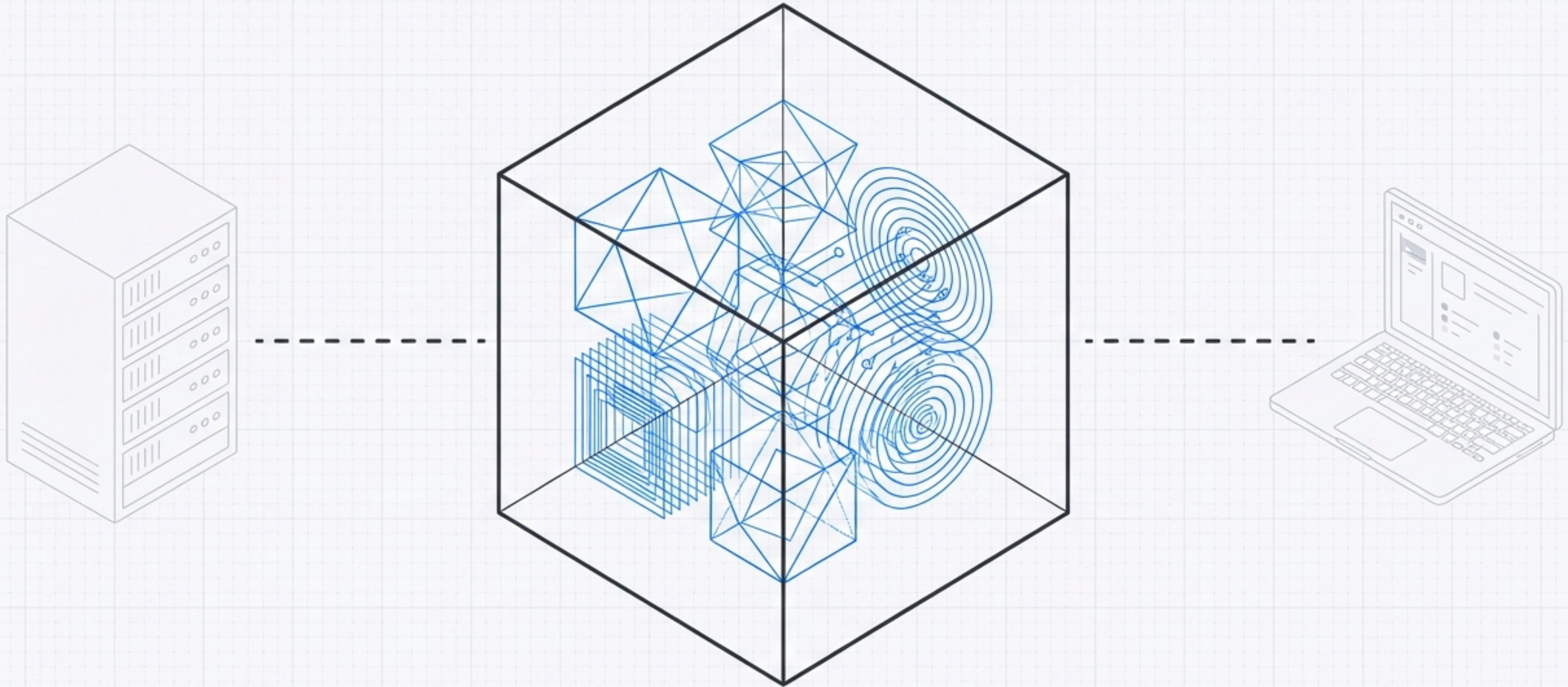


SECURE POD ARCHITECTURE

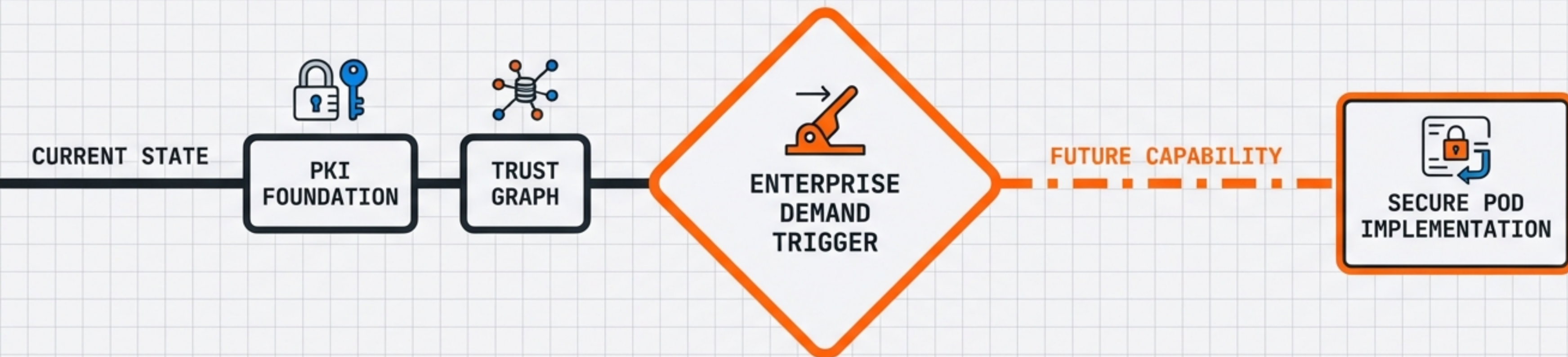
Multi-User Revocable Encryption



Resolving the tension between Privacy, Access,
and Control in Enterprise Data Rooms.

STATUS: RESEARCH (FUTURE CAPABILITY)
VERSION: 0.4.12
DATE: 20 FEB 2026

A Blueprint for Future Capability



STATUS:

Architecture Brief / Research. Not currently on the immediate build roadmap.

THE TRIGGER:

Implementation begins only upon specific customer demand for revocable data rooms with audit trails.

THE FOUNDATION:

Builds directly upon existing PKI, Key Management, and Trust Graph infrastructure.

The Encryption Trilemma

ZERO-KNOWLEDGE

Server never sees plaintext

THE
IMPOSSIBLE
REGION

REVOCABLE

Cut access after
permission

MULTI-USER

One file, many
recipients

In traditional cryptography,
you must choose two.

- To revoke access, you must change the key.
- To change the key for one user without re-encrypting for everyone, you need a central authority.
- If the central authority manages the keys, you lose Zero-Knowledge.

Why Naive Approaches Fail

Every standard approach compromises one critical pillar.

	 Zero-Knowledge	 Multi-User	 Revocable	
Per-User Wrapping	✓	✓	✗	Note: User holds the key.
Group Key	✓	✓	✗	Note: Old content exposed.
Server-Side Re-encryption	✗	✓	✓	Note: Server sees plaintext.

The Fundamental Constraint



**“You can’t
un-know
a key.”**



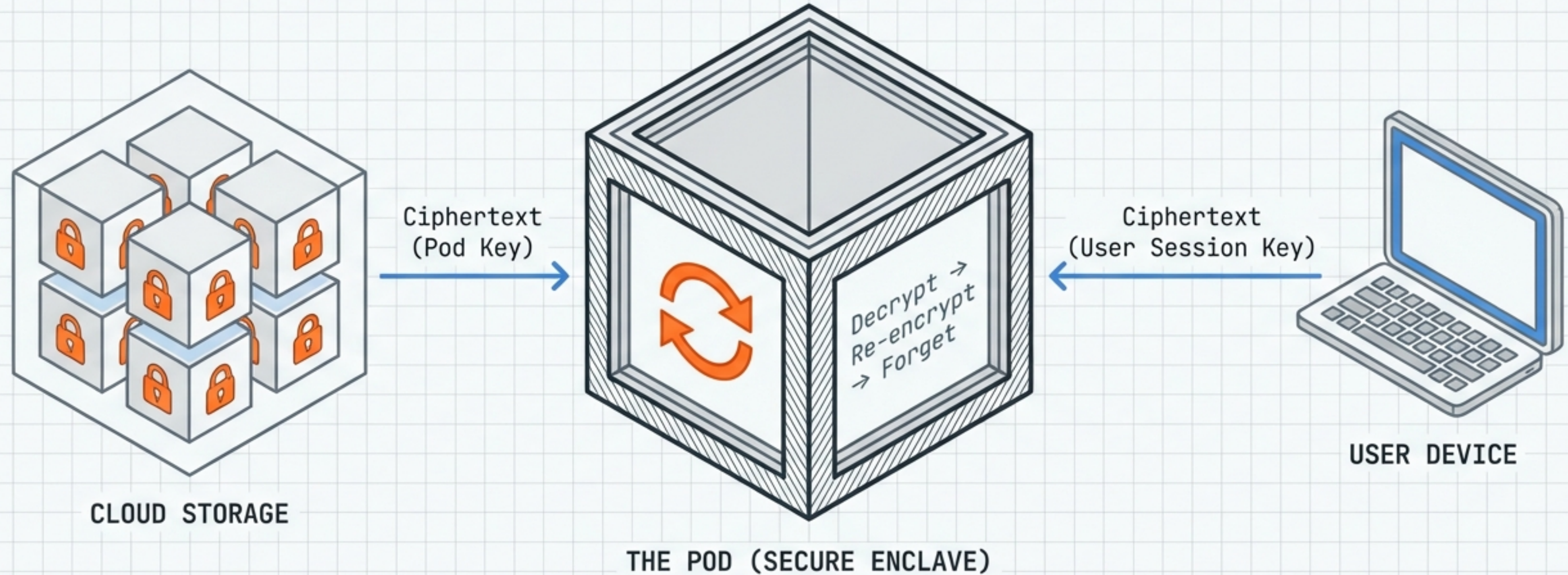
The Reality: We cannot control what a user has already downloaded, screenshotted, or memorized.

The Pivot: We stop trying to control the past.

The New Goal: Ensure the maximum damage from a revoked user is limited strictly to the documents they actually opened. They cannot decrypt the rest of the server.

The Solution: The Secure Pod

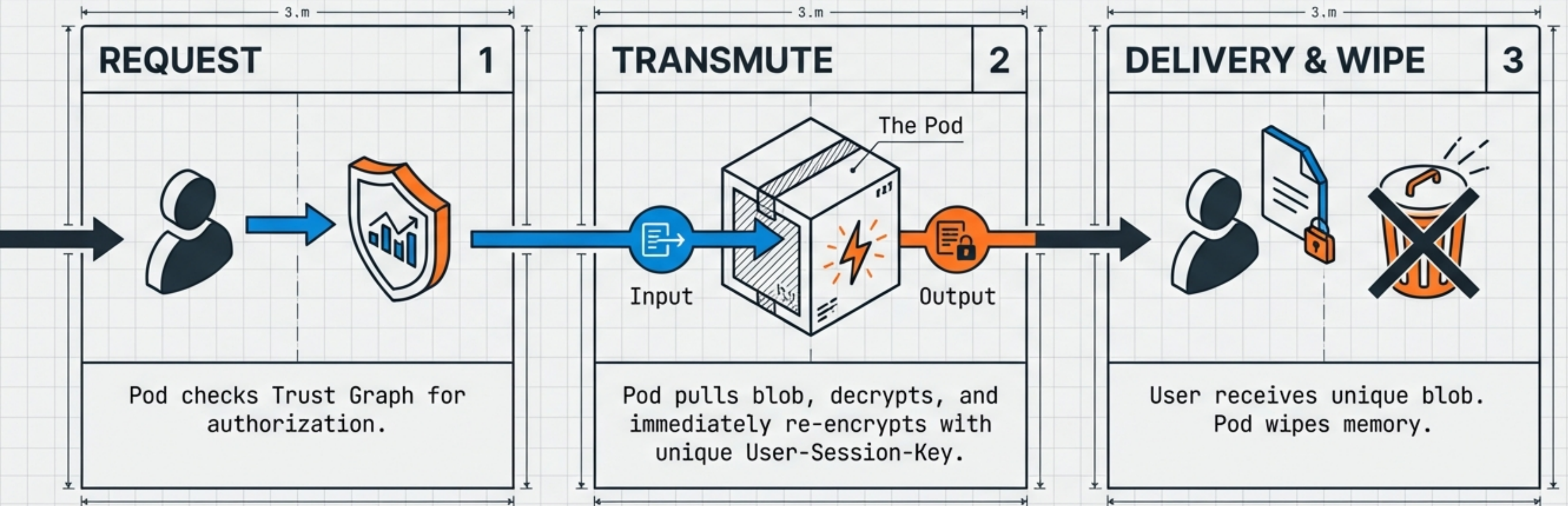
A secure enclave whose only job is to Decrypt → Re-encrypt → Forget.



The Pod acts as a breathless intermediary. It holds the master keys; the user only ever gets a one-time ticket.

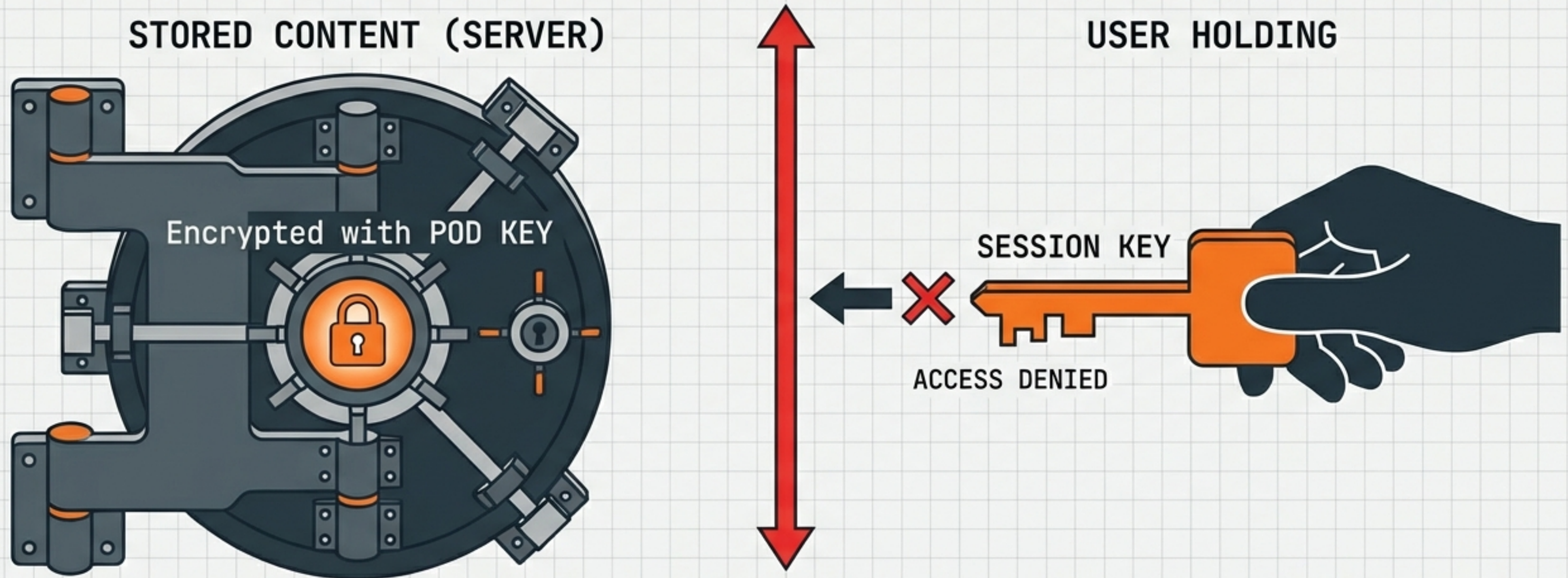
The Access Flow

The Pod sees plaintext momentarily. The Server never does.
The User never holds the master key.



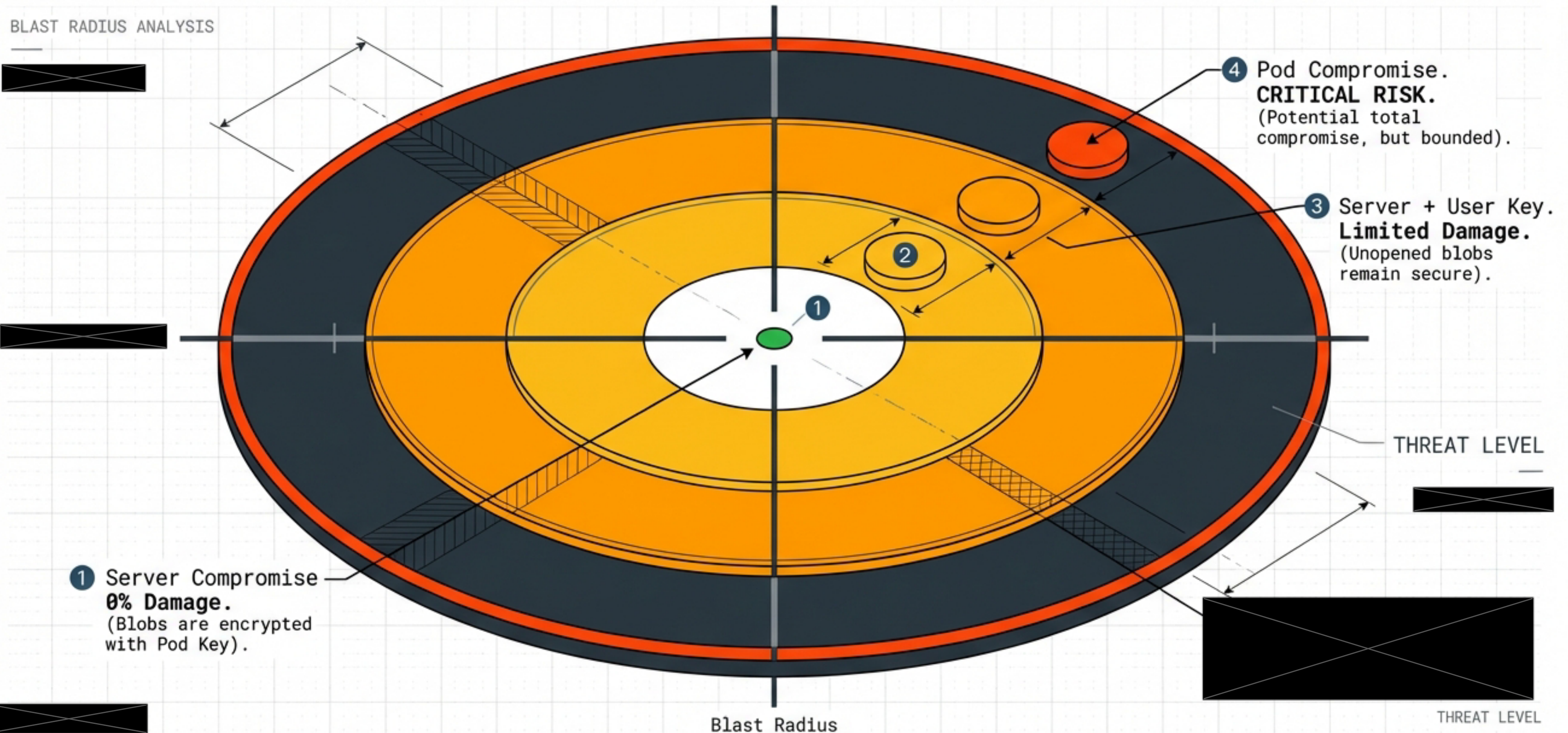
The Golden Rule of Security

The User NEVER holds a key that works on stored content.



If a malicious user dumps the database and leaks their key, they get NOTHING except the specific files they previously opened. The millions of other documents remain locked.

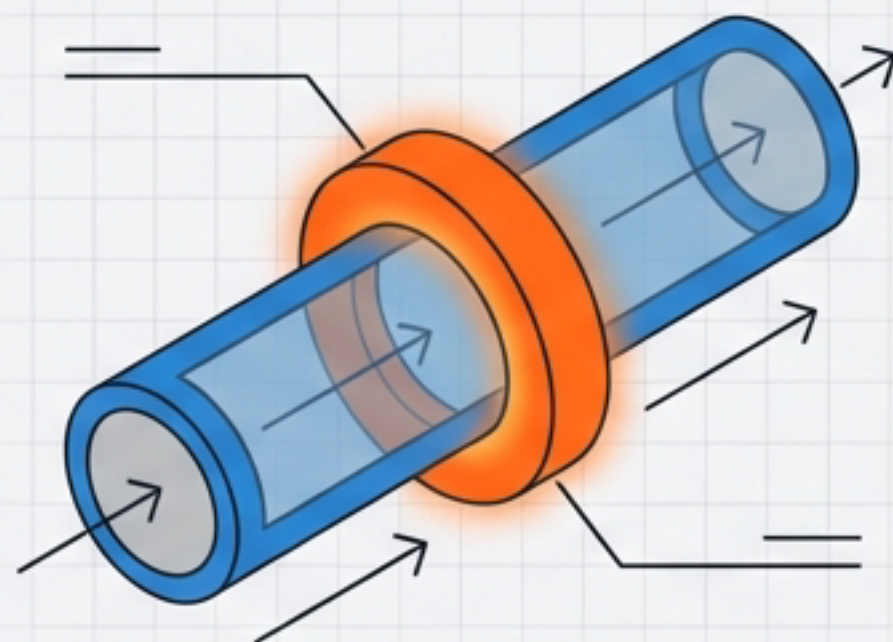
Maximum Damage Model



Containing the Critical Threat

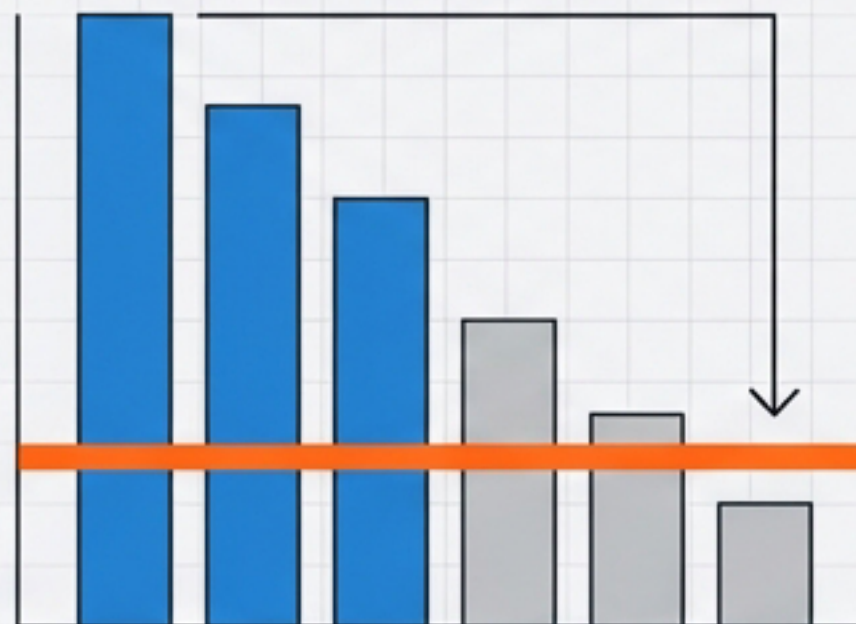
A compromised Pod is an incident, not a catastrophe.

NETWORK PHYSICS



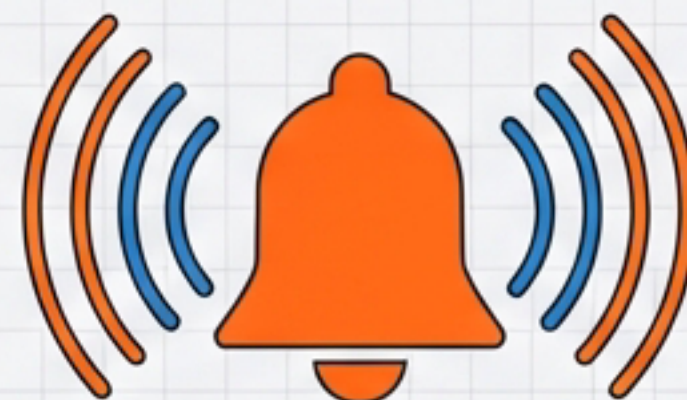
Rate Limits
(Download Volume)

ECONOMIC CIRCUIT BREAKER



Micro-payment Budget
(Cost to Decrypt)

ALERT SYSTEM

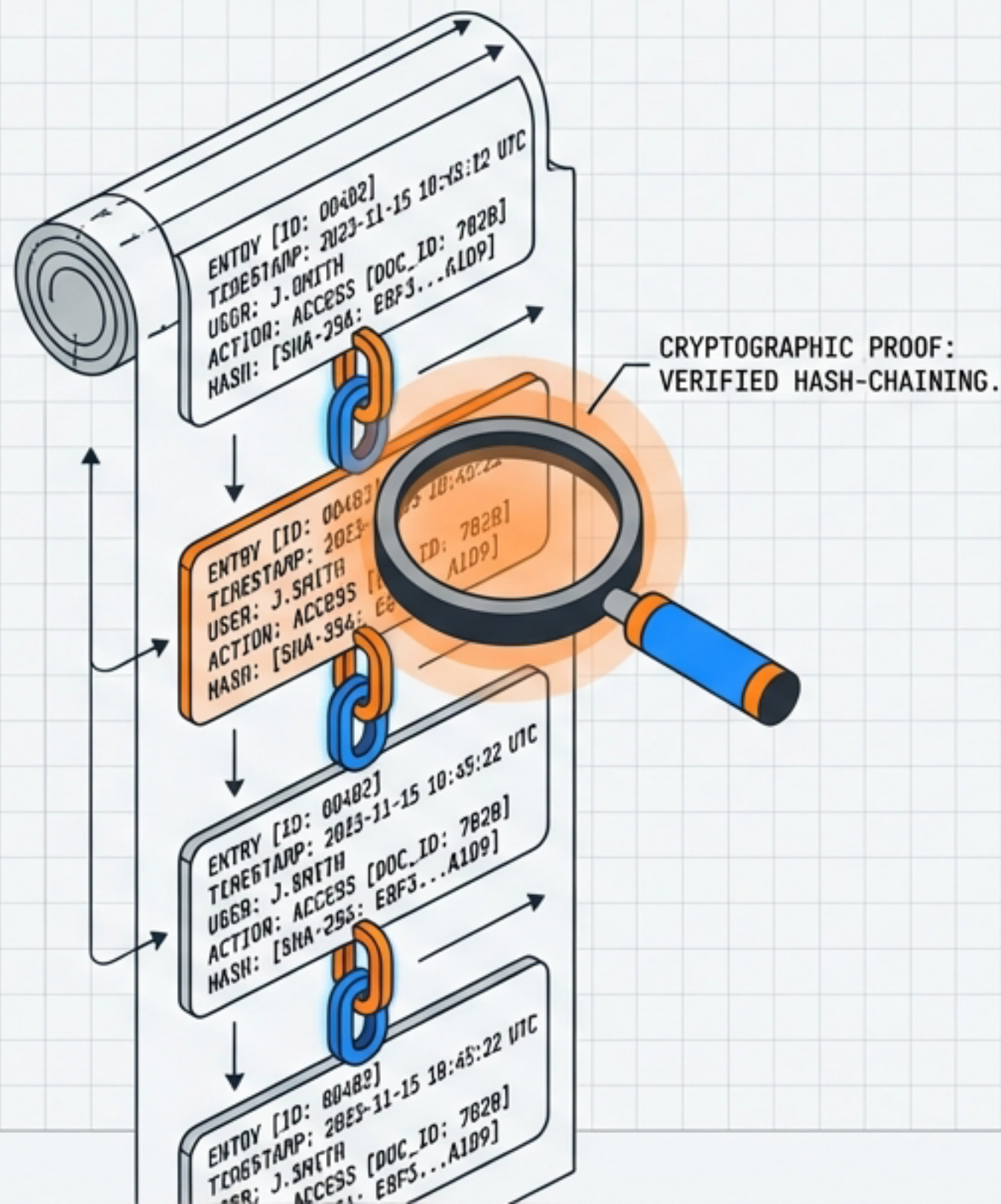


Anomaly Detection

Result: The blast radius is bounded by detection speed and budget, preventing a silent total dump.

The Ultimate Benefit: Provable Audit Trails

Because every access MUST pass through the Pod, the Pod's log is the definitive truth.



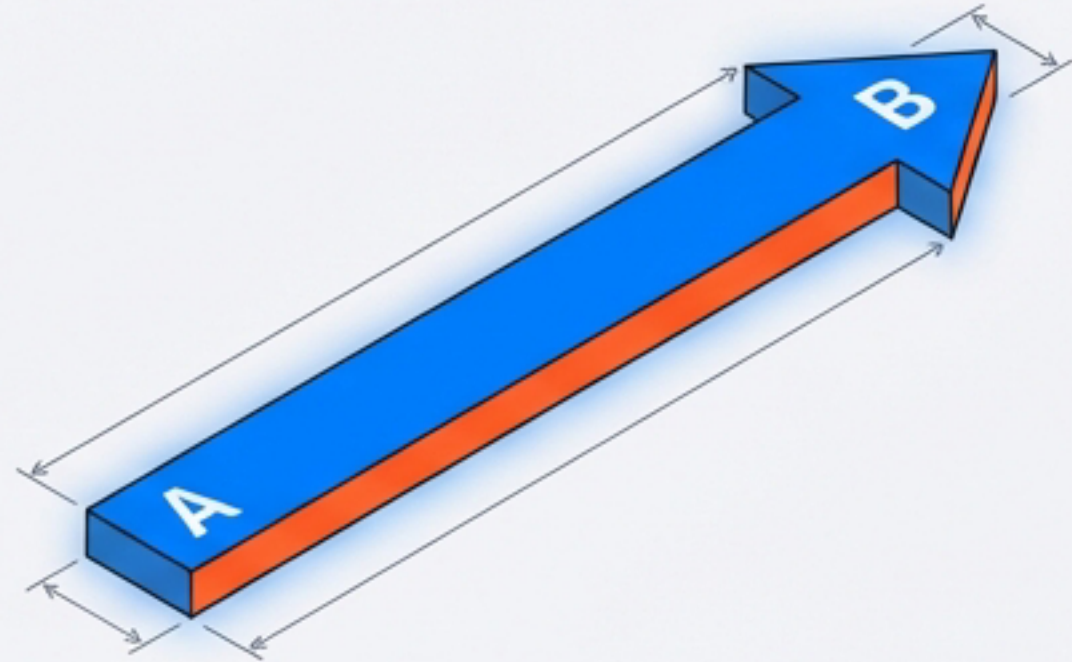
Use Case: Regulatory Compliance (GDPR, HIPAA, Financial).

The Promise: We can cryptographically prove exactly what a user accessed—and crucially, what they did NOT access.

Two Modes of Operation

MODE 1: DIRECT PKI

Current Model



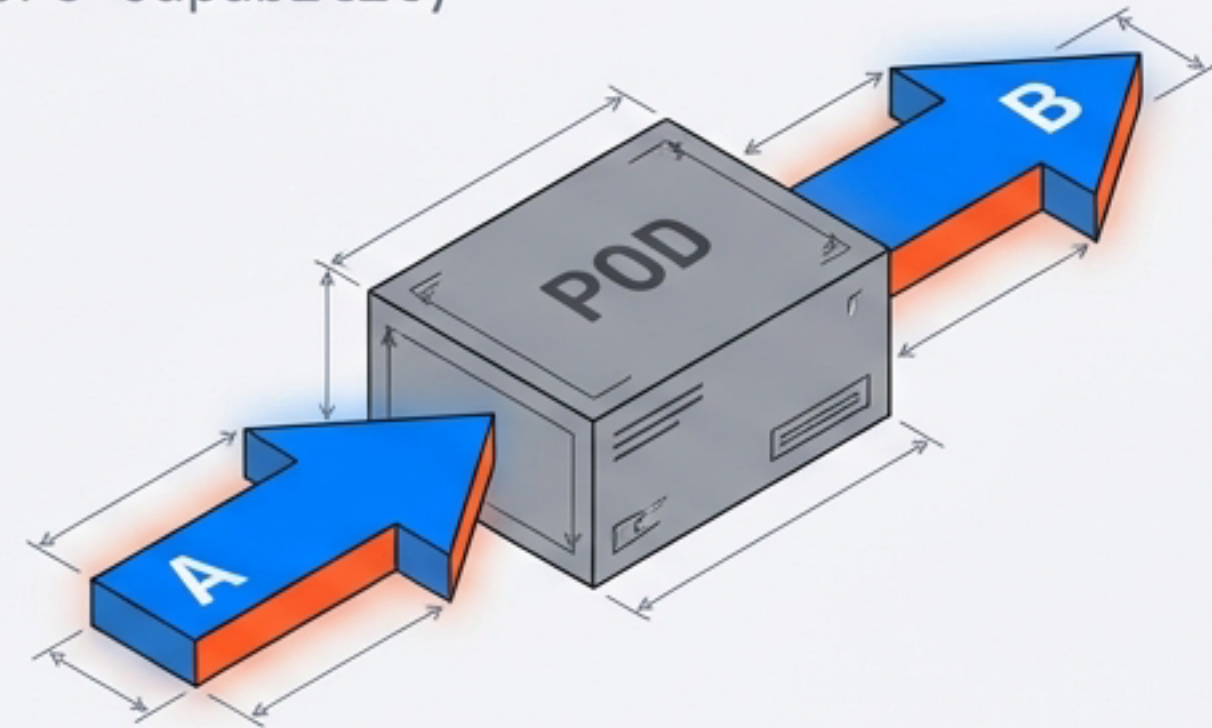
Pros: ✓ Zero-Knowledge (Server & Pod blind)

Cons: No audit trail, No revocation

Use Case: 1:1 Transfers, Personal Messages

MODE 2: POD-MEDIATED

Future Capability



Pros: ✓ Multi-User, ✓ Revocable, ✓ Auditable

Cons: Pod sees plaintext momentarily

Use Case: Data Rooms, Corporate Compliance

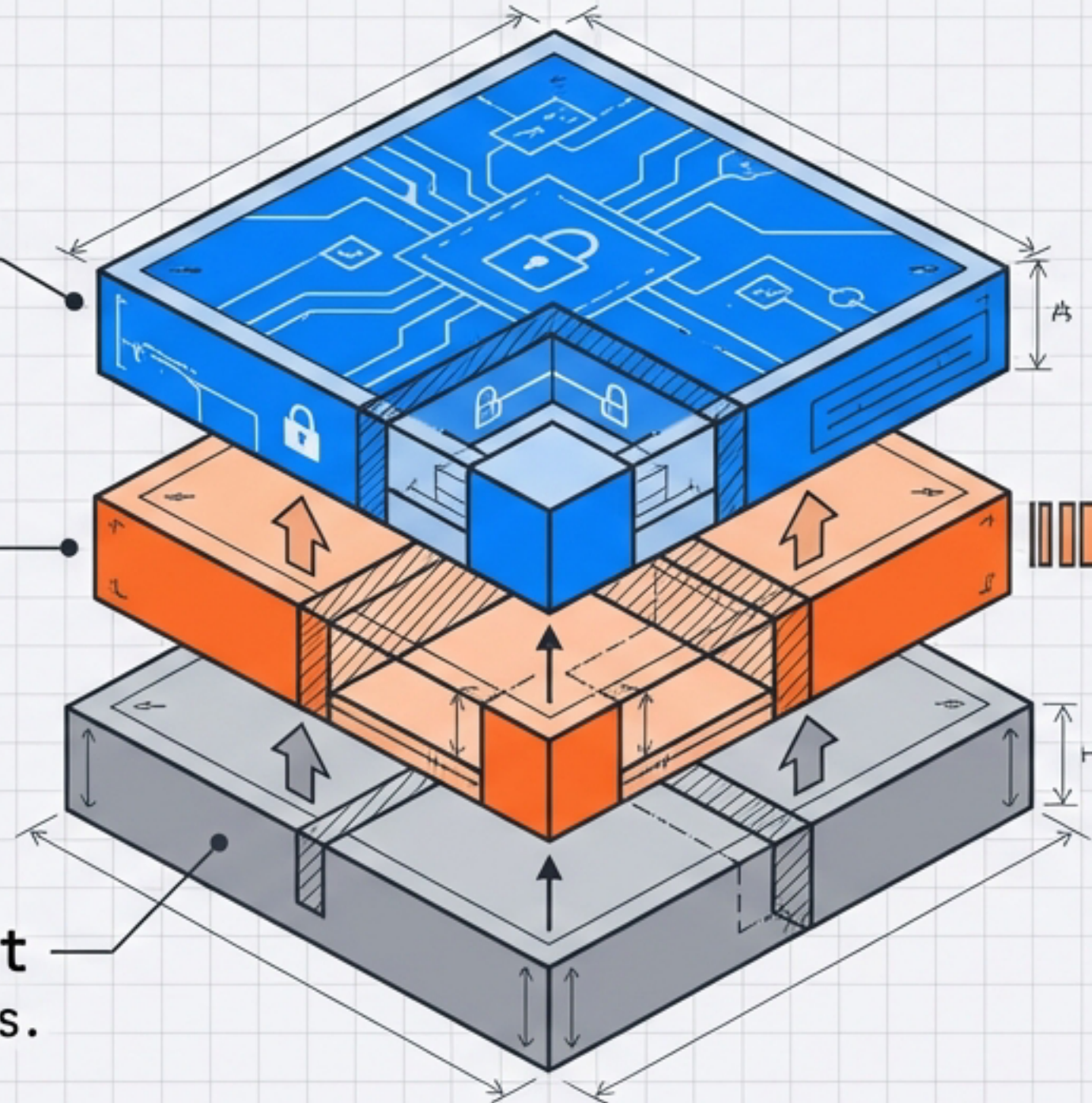
Implementation Architecture

Key Note: Attestation is critical. We must be able to prove the code running in the Pod is unmodified.

HIGH ASSURANCE: CloudHSM
FIPS 140-3 Hardware.

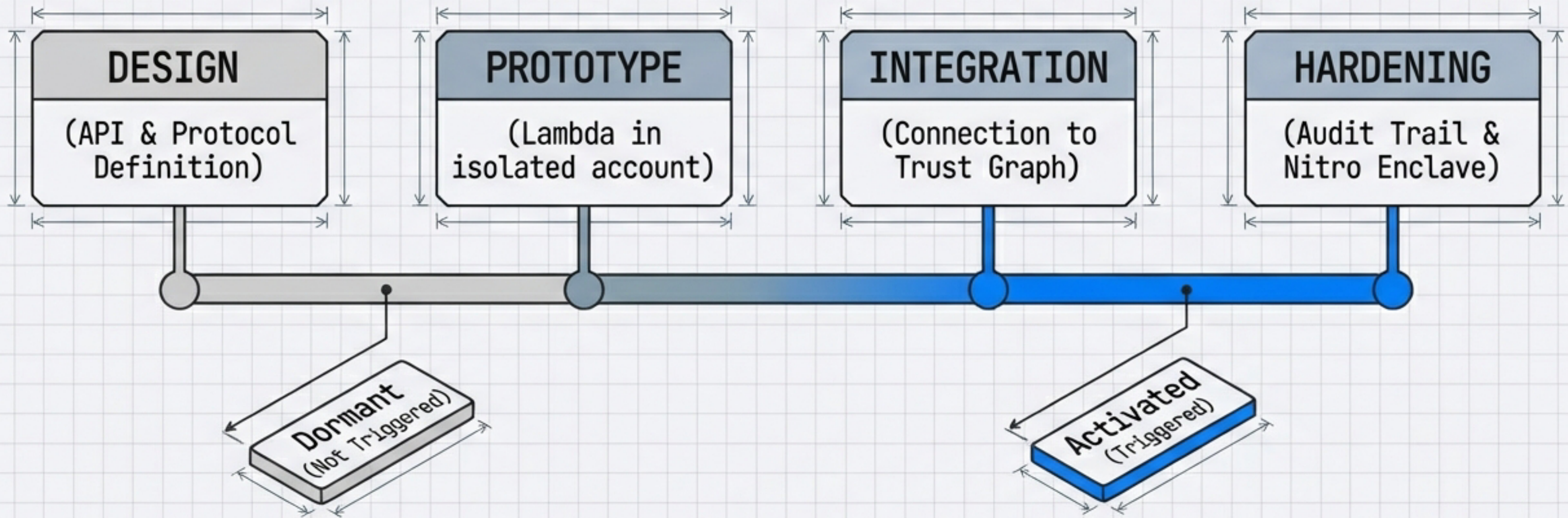
PRODUCTION:
AWS Nitro Enclaves
Hardware isolation, no SSH,
no persistent storage.

MVP: Separate AWS Account
Isolation via strict IAM roles.



Roadmap & Execution Triggers

Neue Haas Grotesk Display, Deep Charcoal



This architecture transforms Revocability from an impossible math problem into a manageable engineering constraint.